

Assignment :- 1

Unit :- I

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Branch :- B.Tech CSE AI & ML

Semester :- IV

Subject :- Deep Learning

① What is deep learning and how is it different from traditional machine learning?

→ Deep learning is a specialized subset of ML, and a branch of AI that utilizes multilayer artificial neural networks to emulate human brains decision making process.

It enables to automatically learn, analyze and make predictions from vast, unstructured datasets such as images video & text.

Eliminating the need for manual feature extraction

Difference:-

- Traditional ML :- Requires manual feature engineering
- Deep learning → learns features automatically using deep neural networks.

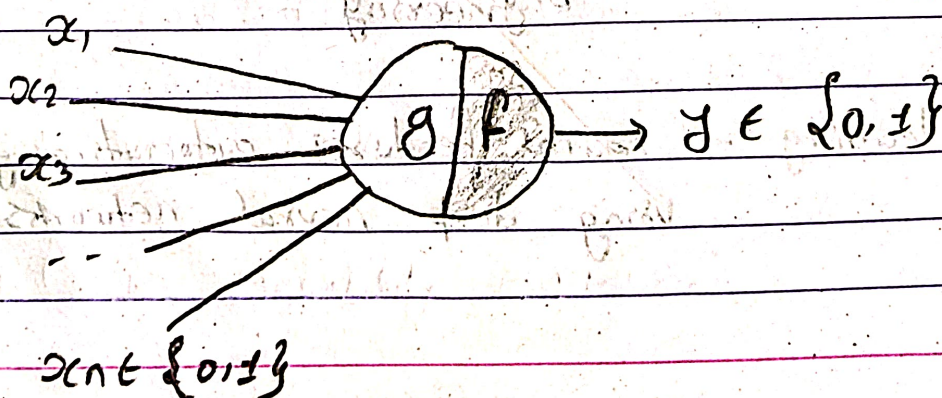
② Briefly explain the historical evolution of deep learning.

- • 1943 - McCulloch & Pitts proposed first artificial neuron model.
- 1958 - Rosenblatt introduced the perceptron.
- 1986 - Backpropagation popularized for training multilayer networks.
- 2006 - Deep belief networks revived interest in deep learning.
- 2012 - AlexNet breakthrough in ImageNet competition accelerated modern deep learning era.

③ What is the McCulloch - Pitts neuron model?

→ A mathematical model of a biological neuron that:

- Takes binary inputs
- Applies weights
- Produces binary output using a threshold function.



④ What are the limitations of the McCulloch-Pitts neuron?

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- only binary inputs and outputs.
 - No learning capability.
 - Cannot solve non-linear problems.
 - Fixed threshold.

⑤ Define a perceptron and mention one of its applications.

→ A perceptron is a single-layer linear binary classifier that maps input features to output using weights and a threshold.

Application: Spam email classification.

⑥ What is meant by a Multilayer Perceptron (MLP)?

→ An MLP is a feedforward neural network with:

- Input layer
- one or more hidden layers
- output layer
- Non-linear activation functions.

⑦ Why are activation functions required in neural networks?

→ They introduce non-linearity, allowing networks to learn complex patterns. Without them, the network becomes just a linear model.
Some activations are:

i) linear

ii) Sigmoid

iii) ReLU

iv) Tanh

⑧ What is a sigmoid neuron and why is the Sigmoid function popular?

→ A Sigmoid neuron uses the Sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Popular because:

- Smooth and differentiable
- Outputs values between 0 and 1.
- Useful for probability modeling.

⑨ what is a feed forward neural network?

→ A neural network where information flows only in one direction - from input to output - with no feedback loops.

⑩ what role do weights and bias play in a neural network?

→ Weights: determine the importance of each input.

Bias: shifts the activation function to improve flexibility.

⑪ what is the representation power of an MLP?

→ An MLP with at least one hidden layer can approximate any continuous function.
(Universal Approximation Theorem).

(12) Why can a single-layer perceptron not solve non-linearly separable problems?

→ A single-layer perceptron not solve non-linearly separable problems because it can only create a linear decision boundary.

(13) What is learning in the context of neural networks?

→ Learning is the process of adjusting weights and bias to minimize prediction error.

(14) Define back propagation in simple terms.

→ Backpropagation is a method that:

- calculates error at output
- sends it backward through the network
- updates weights using gradients

Q3) What is the role of error / loss function in training a neural network?

→ It measures how far predictions are from actual values and guides weight updates.

Long Answer Question:

Q1) Explain the history and development of deep learning, highlighting key milestones and breakthroughs.

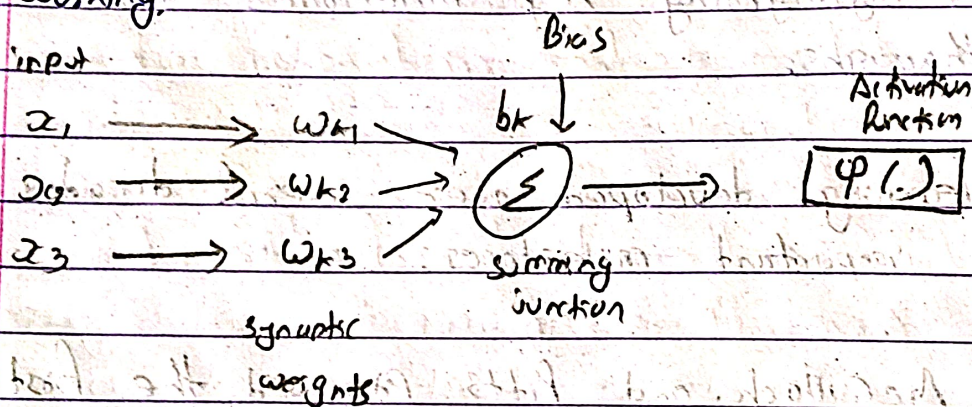
→ Deep learning developed over several decades through important milestones:

- 1943 - McCulloch and Pitts proposed the first artificial neuron model.
- 1958 - Rosenblatt introduced the perceptron for binary classification.
- 1969 - Minsky showed limitations of perceptrons (XOR problem), slowing research.
- 1986 - Backpropagation algorithm enabled training of multilayer networks.
- 2006 - Deep belief networks revived deep learning research.

- 2012 - AlexNet achieved major success in image recognition using GPUs.

With large datasets and powerful hardware, deep learning became dominant in AI applications like computer vision and NLP.

- ② Describe the McCulloch - Pitts neuron model with a neat diagram and explain its working.



The McCulloch - Pitts neuron is a mathematical model of a biological neuron.

Components:

- Binary inputs
- weights
- Summation unit
- Threshold function

Working :-

1) compute weighted sum :

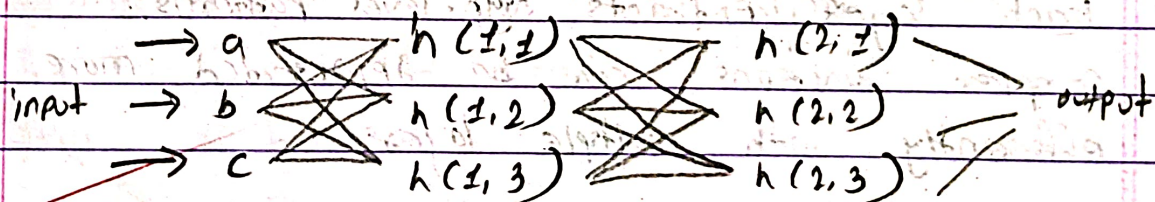
$$S = \sum w_i x_i$$

output = 1 if $S \geq \theta$, otherwise 0.

It can perform simple logical operations like AND and OR.

③ What is a multilayer perceptron (MLP)? Explain its architecture and components in detail.

→ A multilayer perceptron (MLP) is a feedforward neural network with one or more hidden layers.



Architecture

- Input layer
- Hidden layer
- Output layer

Components

- weights
- Bias
- Activation functions
- Loss function

Each neuron computes:

$$z = \sum w_i x_i + b$$

$a = \sigma(z)$

MLP can solve complex non-linear problems.

④ Explain the representation power of MLPs and discuss why deeper networks are more expressive.

→ An MLP with at least one hidden layer can approximate any continuous functions.

Universal approximation theory

An MLP with:

- At least one hidden layer
- A sufficient no. of neurons
- A non-linear activation function

Deeper networks are more expressive because:

- They learn hierarchical features.
- Each layer extracts high-level patterns.
- Complex functions can be represented more efficiently with multiple layers.

Thus, deeper networks require fewer neurons compared to very wide shallow networks.

⑤ Discuss the Sigmoid activation function. Explain its mathematical properties, advantages, and limitations.

→ The Sigmoid function is defined as:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Properties:-

- Smooth and differentiable
- output range between 0 and 1

Advantages:-

- Suitable for binary classification
- Interpretable as probability

Limitations:-

- Vanishing gradient problem
- Not zero-centered
- Slower convergence in deep networks

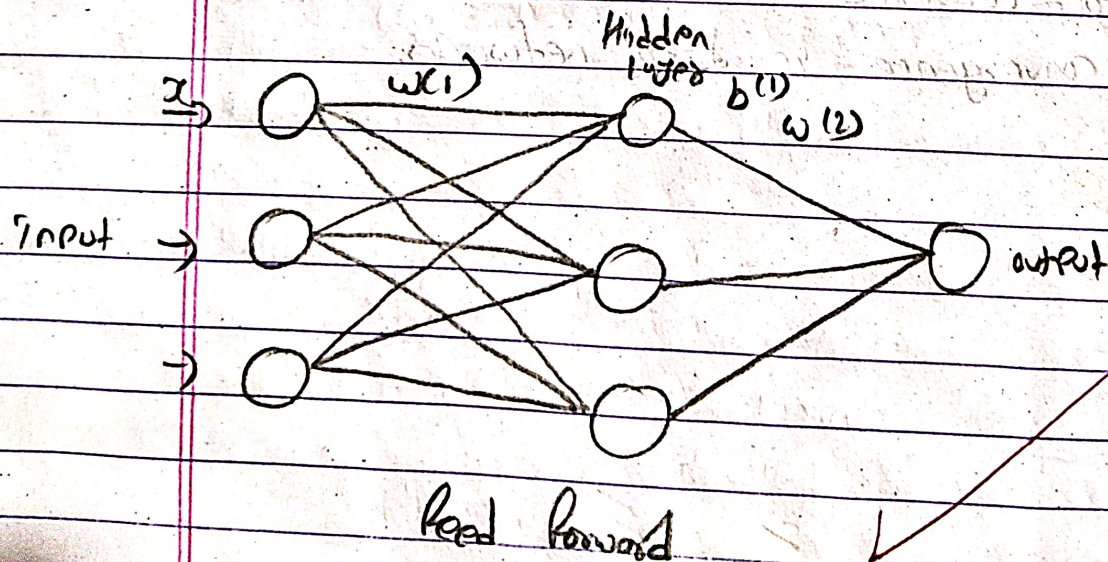
⑥ Explain the architecture and working of a feed forward neural network with an example.

→ A feedforward neural network allows data to move only in one direction - from input to output.

Working:

- i) Input features enter input layers.
- ii) Hidden layers compute weighted sums and apply activation functions.
- iii) Output layer produces prediction.

ex:- In house price prediction, inputs like size and location are processed to predict price.



⑦ What is the back propagation algorithm?
Explain its steps clearly with mathematical intuition.

→ Backpropagation is an algorithm used to train neural networks by minimizing errors.
Steps:

- i) Initialize weights randomly
- ii) Perform forward pass to compute output
- iii) Compute loss using loss function
- iv) Calculate gradients using chain rule
- v) Update weights:

$$w = w - \eta \frac{\partial L}{\partial w}$$

Here,

η is the learning rate.

- It reduces prediction error iteratively.

⑧ Explain how errors are propagated backward in a neural network during training.

→ After computing loss at the output layer:

- The error is calculated as difference between predicted and actual values.
- Using derivatives of activation functions, error is distributed backward.
- Each weight is updated based on its contribution to the error.

This process uses the chain rule to efficiently compute gradients layer by layer.

⑨ Compare single-layer perceptron and multilayer perceptron in terms of structure and learning capability.

Feature	Single-layer perceptron	multilayer perceptron
Hidden layers	None	one or more
Decision Boundary	Linear	Non-linear
XOR Problem	cannot solve	can solve
Learning power	Limited	High

MLP is more powerful due to hidden layers and non-linearity.

Q7) Explain how a neural network learns from data, starting from data, starting from initialization to convergence.

→ Learning continues until the loss stabilizes or reaches minimum values. This point is called convergence.

- i) Initialize weights randomly.
 - ii) Perform forward propagation
 - iii) Compute loss
 - iv) Apply backpropagation to compute gradients.
 - v) update weights using gradient descent.
 - vi) Repeat for many epochs.
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